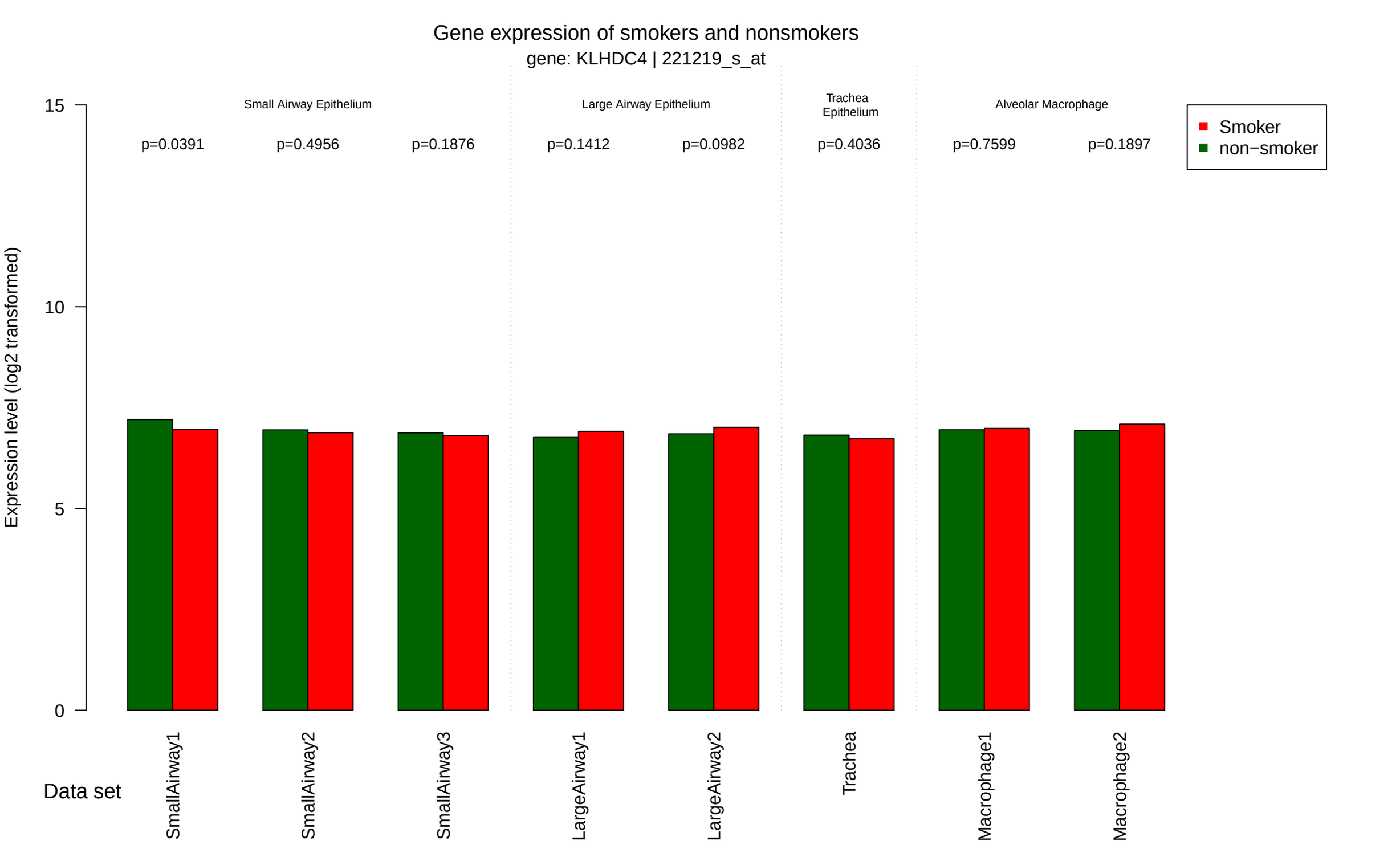
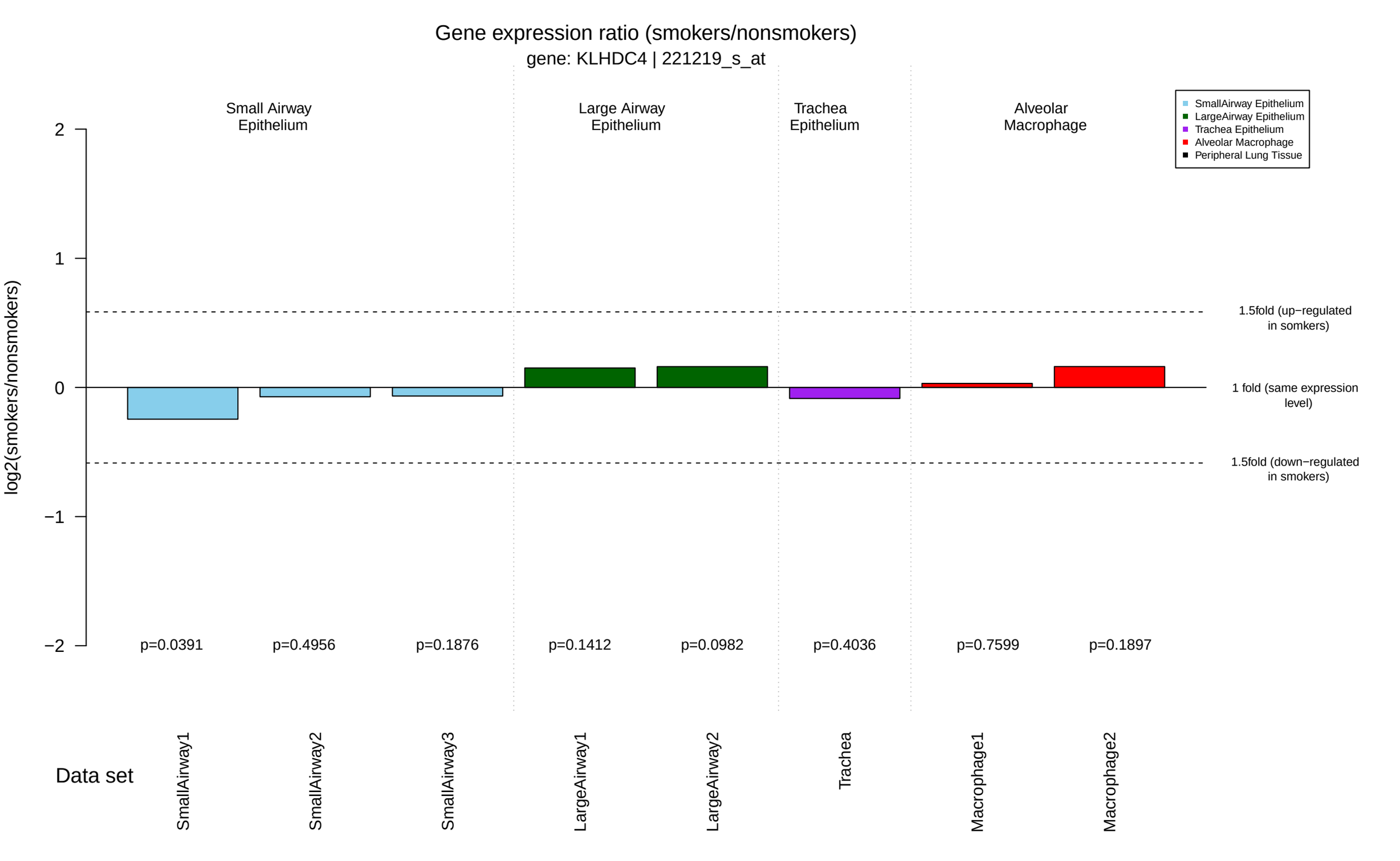
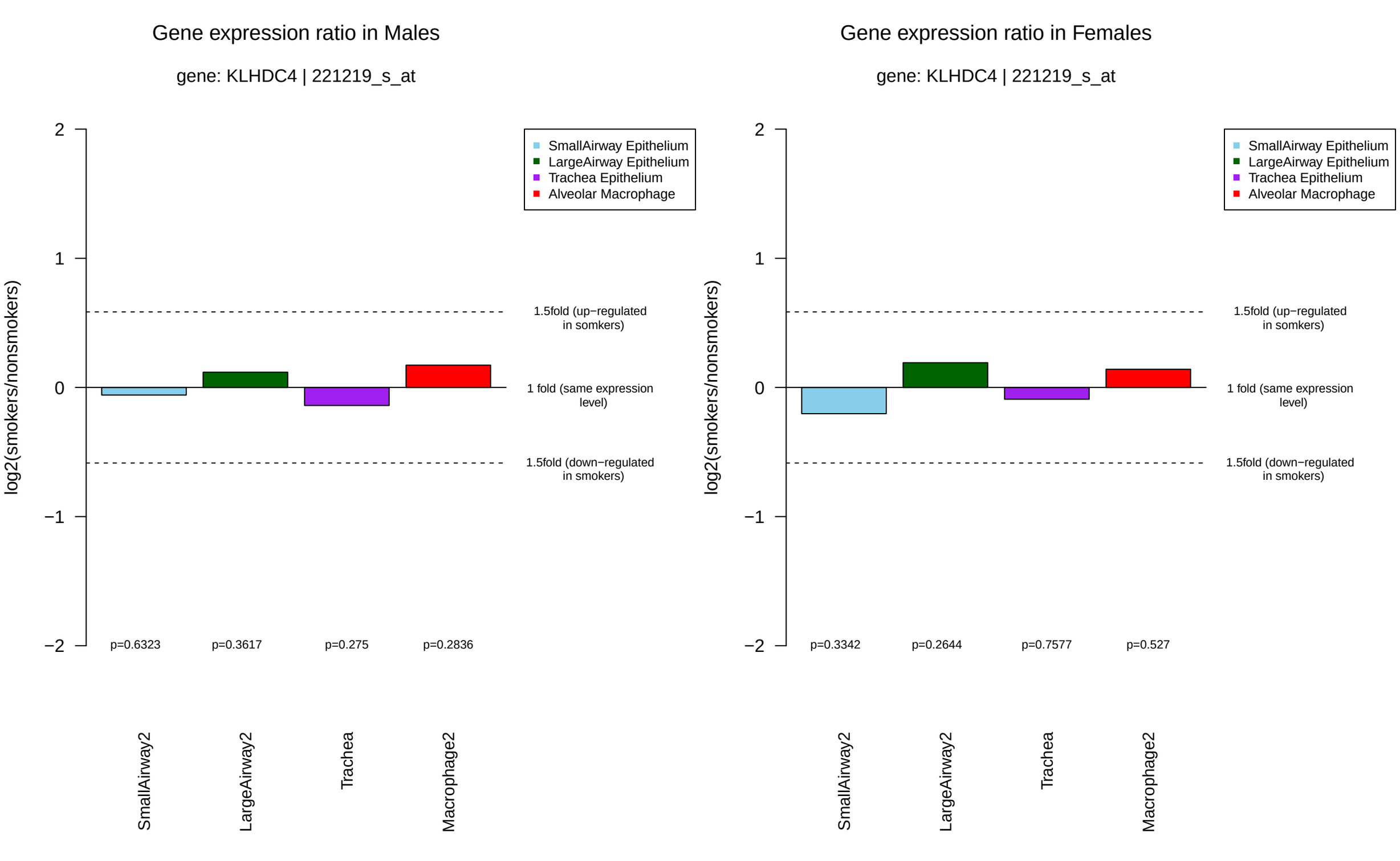
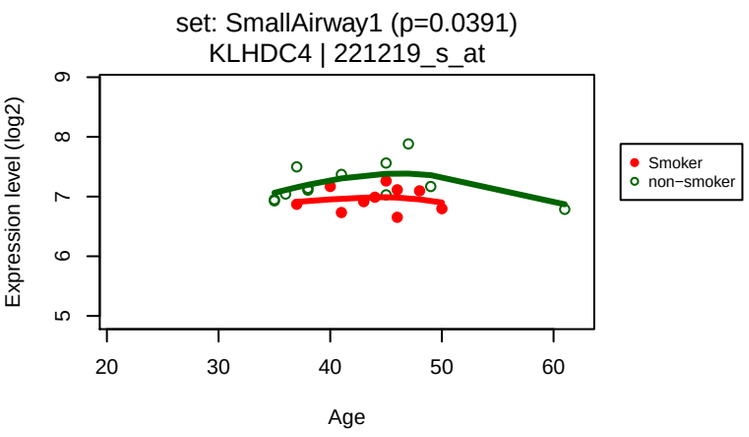
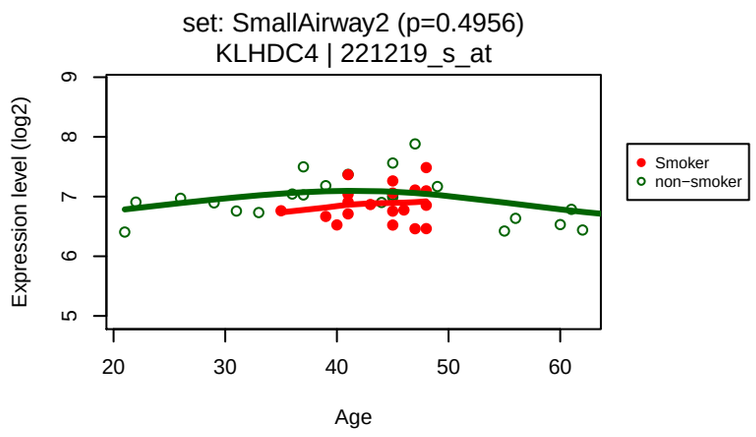
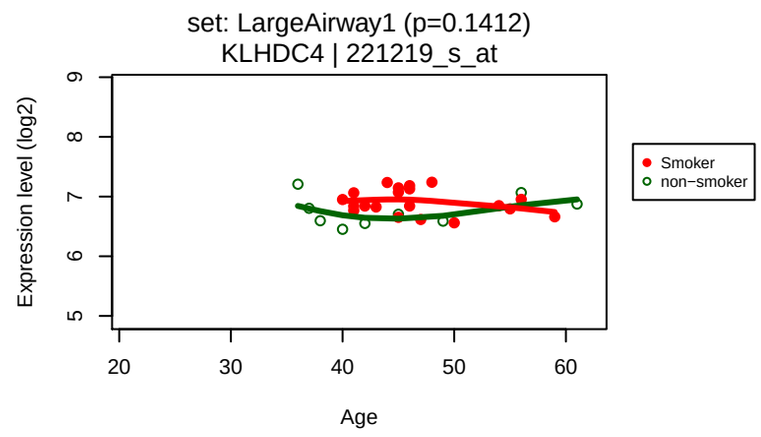

Gene expression of smokers and nonsmokers
gene: KLHDC4 | 221219_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0391
p=0.4956
p=0.1876
p=0.1412
p=0.0982
p=0.4036
p=0.7599
p=0.1897
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

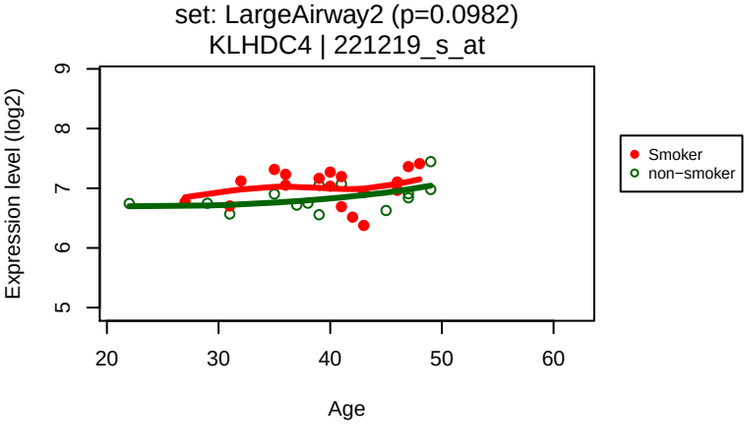

Gene expression ratio (smokers/nonsmokers)
gene: KLHDC4 | 221219_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0391
p=0.4956
p=0.1876
p=0.1412
p=0.0982
p=0.4036
p=0.7599
p=0.1897
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

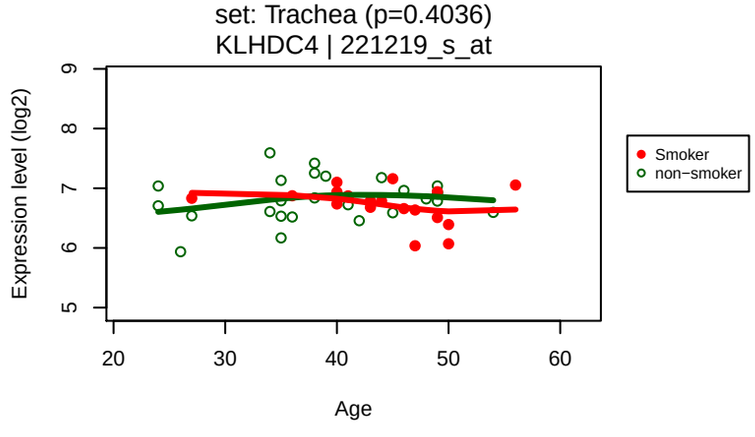

Gene expression ratio in Males
gene: KLHDC4 | 221219_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6323
p=0.3617
p=0.275
p=0.2836
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KLHDC4 | 221219_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3342
p=0.2644
p=0.7577
p=0.527
SmallAirway2
LargeAirway2
Trachea
Macrophage2

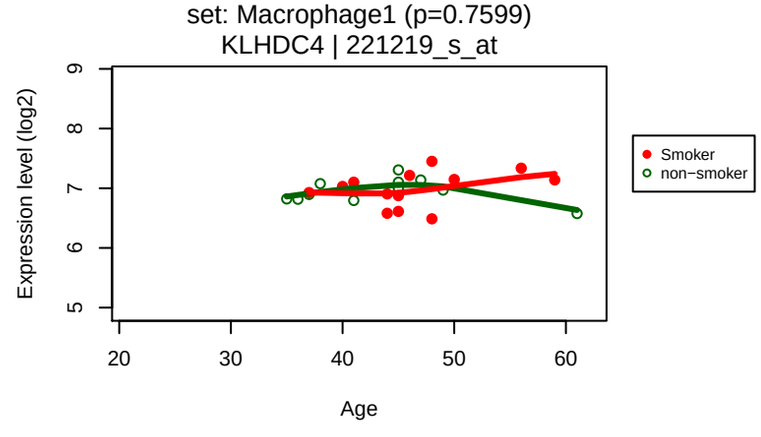

set: SmallAirway1 (p=0.0391)
KLHDC4 | 221219_s_at
Expression level (log2)
Age
Smoker
non−smoker

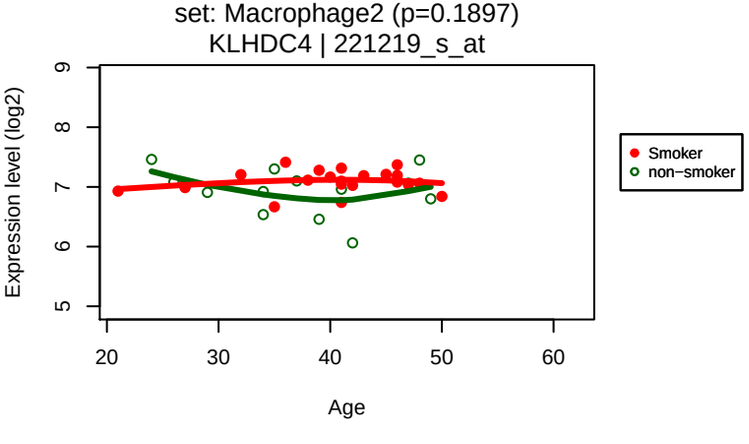

set: SmallAirway2 (p=0.4956)
KLHDC4 | 221219_s_at
Expression level (log2)
Age
Smoker
non−smoker

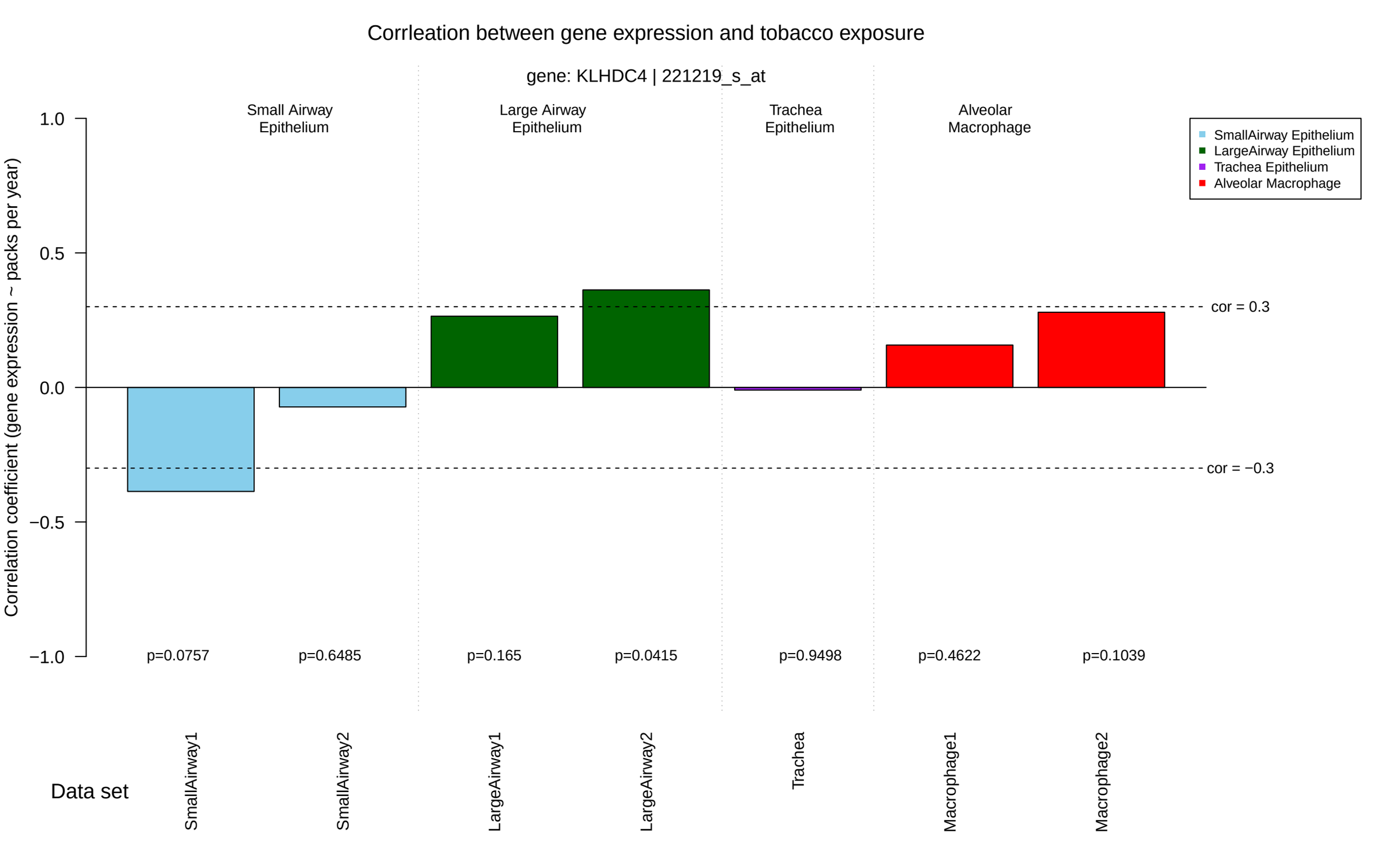

set: LargeAirway1 (p=0.1412)
KLHDC4 | 221219_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0982)
KLHDC4 | 221219_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4036)
KLHDC4 | 221219_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7599)
KLHDC4 | 221219_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1897)
KLHDC4 | 221219_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KLHDC4 | 221219_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0757
p=0.6485
p=0.165
p=0.0415
p=0.9498
p=0.4622
p=0.1039
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2